We proved that if $f:[a,b]\to\mathbb{R}$ is integrable, then if we choose a partition P of $[a,b]$ consisting of n equally-spaced points and any tagging $\tau=\{x_i^*\}$ of P, then

$$\lim_{n\to\infty}\sum_{i=1}^{n} f(x_i^*)\Delta x=\int_a^b f(x)dx.$$

Let's verify this for some simple functions.

Recall from last time the theorem

Theorem 0.1 (Sum Properties)**.** *If c is any constant (that is, that does not depend on i), then*

(a) $\sum_{i=m}^{n} ca_i = c\sum_{i=m}^{n} a_i$

(b) $\sum_{i=m}^{n}(a_i+b_i)=\sum_{i=m}^{n} a_i+\sum_{i=m}^{n} b_i$

(c) $\sum_{i=m}^{n}(a_i-b_i)=\sum_{i=m}^{n} a_i-\sum_{i=m}^{n} b_i$

and the sum formulas

- $\displaystyle\sum_{i=1}^{n} 1 = n$
- $\displaystyle\sum_{i=1}^{n} i = \frac{n(n+1)}{2}$
- $\displaystyle\sum_{i=1}^{n} i^2 = \frac{n(n+1)(2n+1)}{6}$
- $\displaystyle\sum_{i=1}^{n} i^3 = \left[\frac{n(n+1)}{2}\right]^2$

1. Express each of the following integrals as limits of Riemann sums, using the right-endpoint rule ($x_i^*=x_i=a+i\Delta x$). Do not attempt to evaluate.

   (a) $\displaystyle\int_0^{\pi}\sin(x)dx$

   (b) $\displaystyle\int_0^{4}\frac{dx}{x^4+1}$

   (c) $\displaystyle\int_2^{7}\frac{dx}{x^2}$

2. The following limits can represent integrals. Determine an integral for each.

   (a) $\lim_{n\to\infty} \sum_{i=1}^{n} e^{\left(\frac{i}{n}\right)} \cdot \frac{1}{n}$

   (b) $\lim_{n\to\infty} \sum_{i=1}^{n} \left(\frac{10i}{n}\right)^{\frac{1}{3}} \cdot \frac{5}{n}$

   (c) $\lim_{n\to\infty} \sum_{i=1}^{n} 3 \cdot \left(4 + \frac{4j}{n}\right)^{2} \cdot \frac{4}{n}$

   (d) $\lim_{n\to\infty} \frac{2}{n} \cdot \sum_{i=1}^{n} \frac{1}{\sqrt{1+\frac{k}{n}}}$

3. Evaluate $\int_{-1}^{5}(2x+3)$ as the limit of a Riemann sum using the right-endpoint rule.

4. Evaluate $\int_0^3 (x^3 - 6x)dx$ as the limit of a Riemann sum using the right-endpoint rule.